Linear Algebra Review

CSE0613409

Machine Vision and Motion Analysis

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What are images? (in Matlab)

- Matlab treats images as matrices of numbers
- To proceed, let's talk very briefly about how images are formed

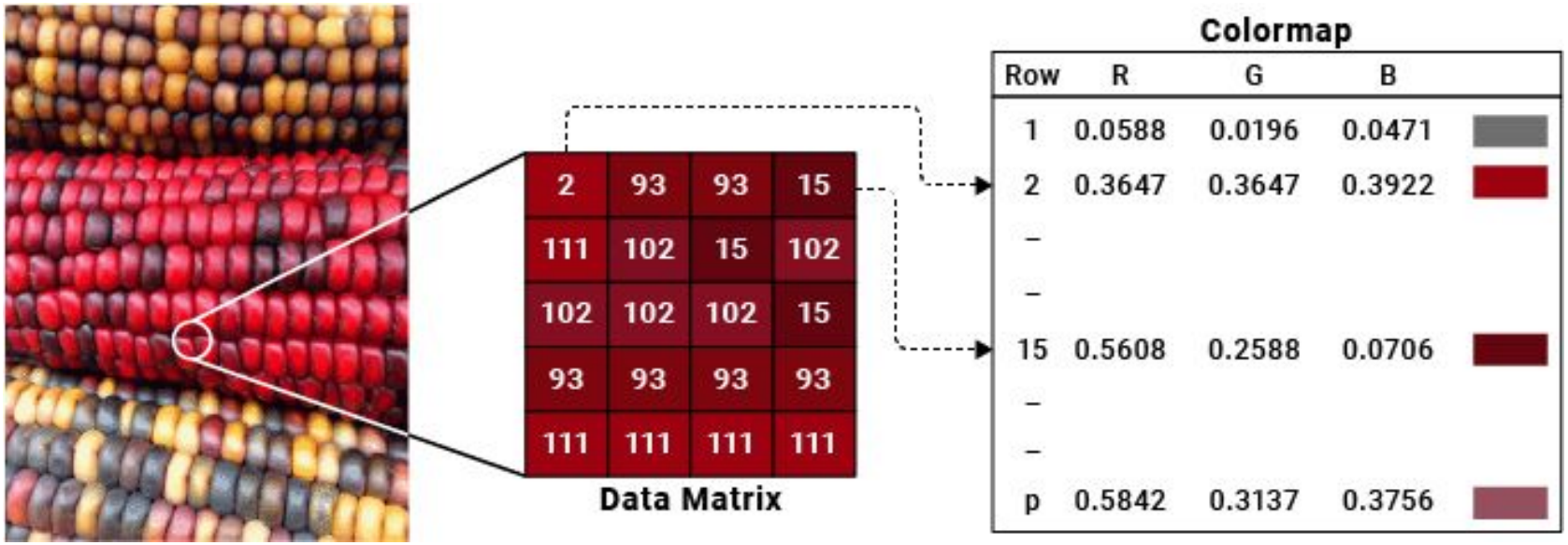
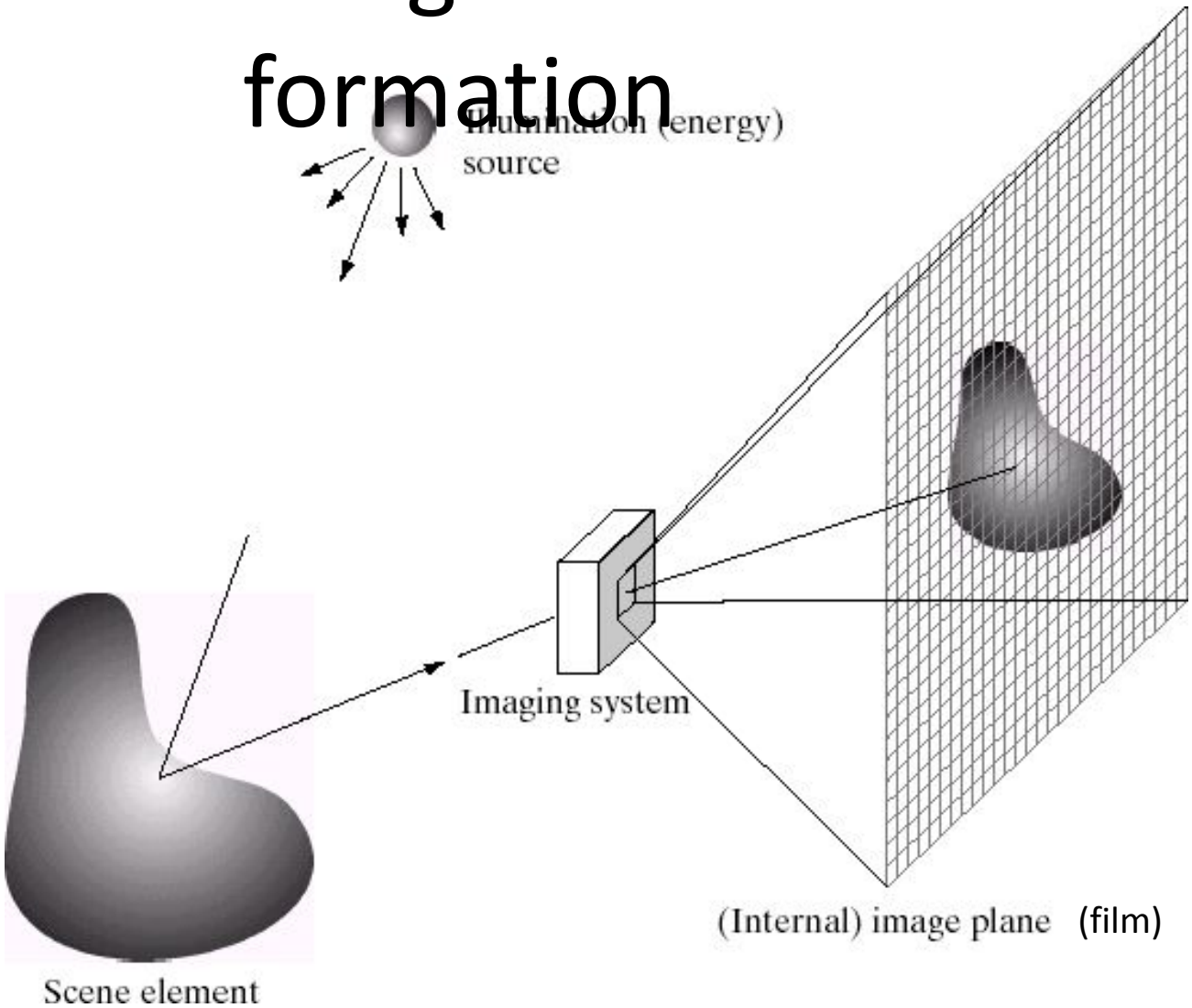


Image formation



Slide credit: Derek Hoiem

Digital camera

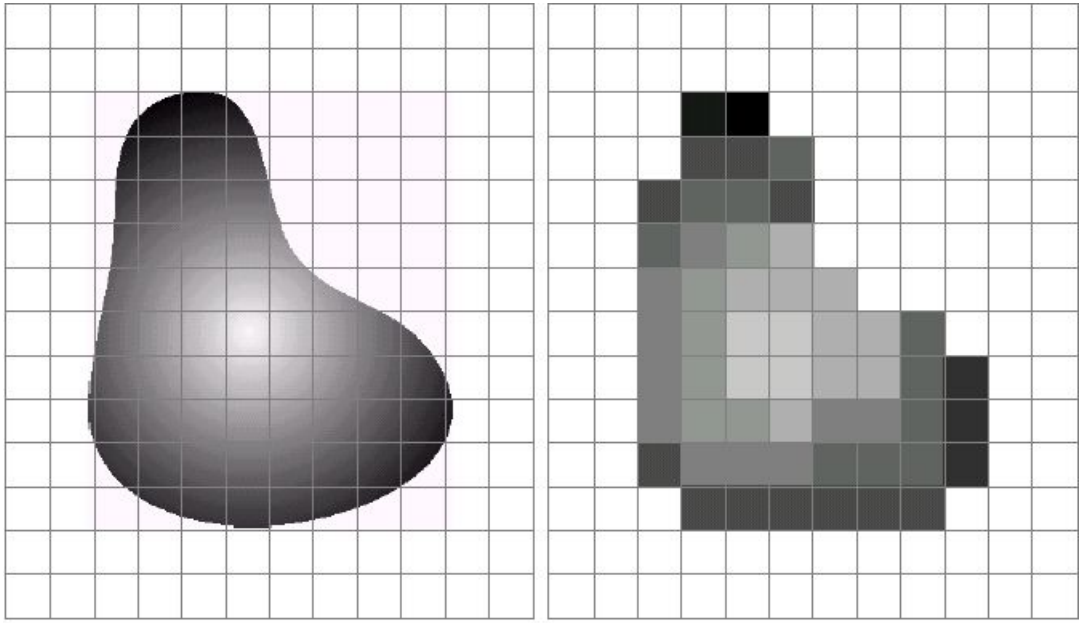
A digital camera replaces film with a sensor array

- Each cell in the array is light-sensitive diode that converts photons to electrons

http://electronics.howstuffworks.com/cameras-photography/digital/digital-camera.htm

Slide by Steve Seitz

Digital images



a b

FIGURE 2.17 (a) Continuos image projected onto a sensor array. (b) Result of image sampling and quantization.

- **Sample** the 2D space on a regular grid
- **Quantize** each sample (round to nearest integer)

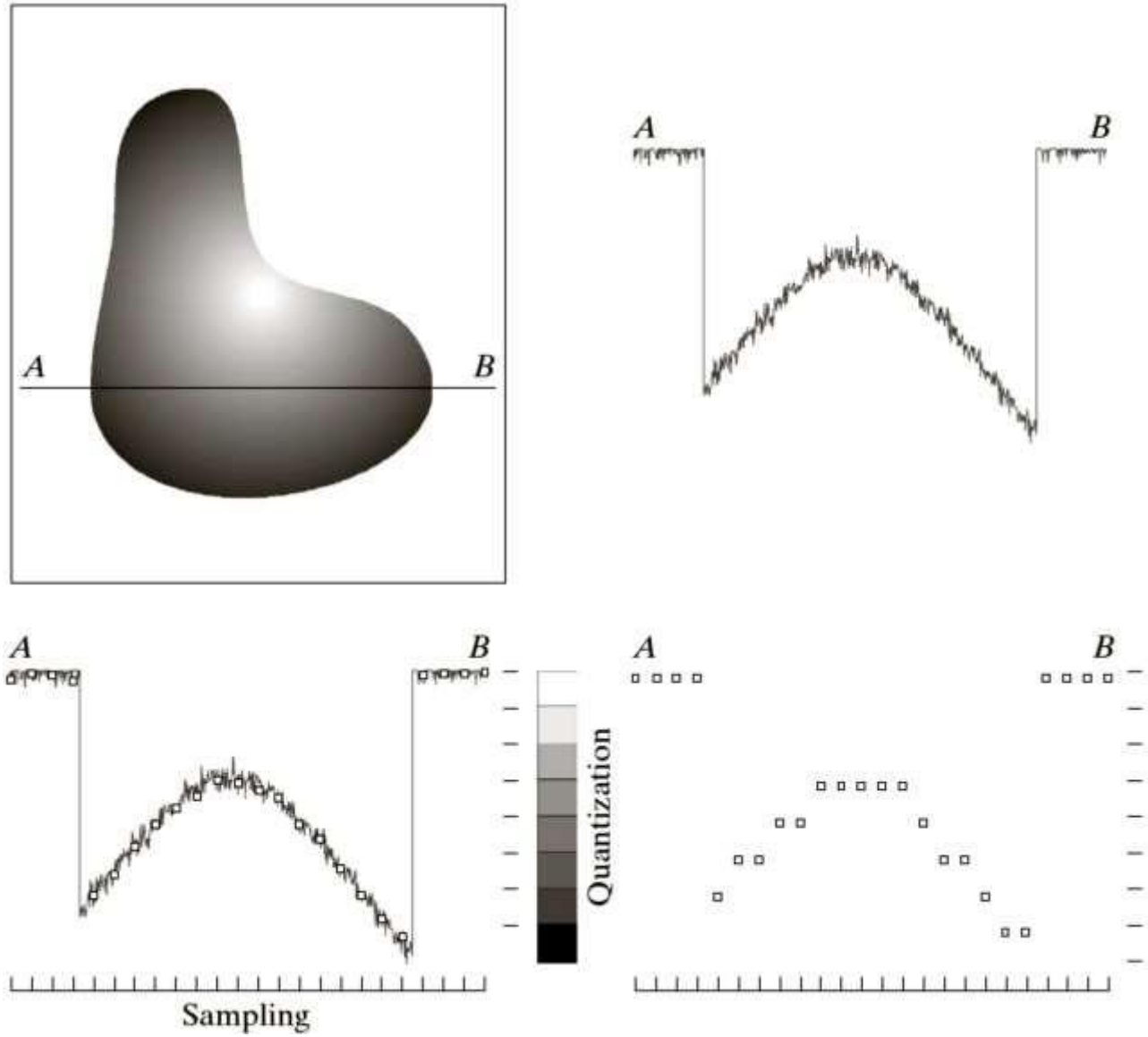
Slide credit: Derek Hoiem, Steve Seitz

Image sampling and quantization

There are numerous ways to acquire images but our objective to *generate digital images from sensed data*. To create digital image, we need to convert the continuous sensed data into digital form. This involves two processes: ***sampling*** and ***quantization***.

- The basic idea behind sampling and quantization is illustrated in Fig. 2.16. Fig. 2.16 (a) shows a continuous image, f(x, y), that we want to convert digital form. *An image may be continuous* with respect to *the x- and y- coordinates and also in amplitude*. To convert it to digital form, we have to sample the function in both coordinates and in amplitude.
- Digitizing the coordinate values is called ***sampling***.
- Digitizing the amplitude values is called the ***quantization.***

Image Sampling & Quantization



a b
c d

FIGURE 2.16 Generating a digital image. (a) Continuous image. (b) A scan line from *A* to *B* in the continuous image, used to illustrate the concepts of sampling and quantization. (c) Sampling and quantization. (d) Digital scan line.

Cont.

The one dimensional function shown in 2.16 (b) is a plot of amplitude (gray level) values of the continuous image along the line segment AB in fig. 2.16 (a).

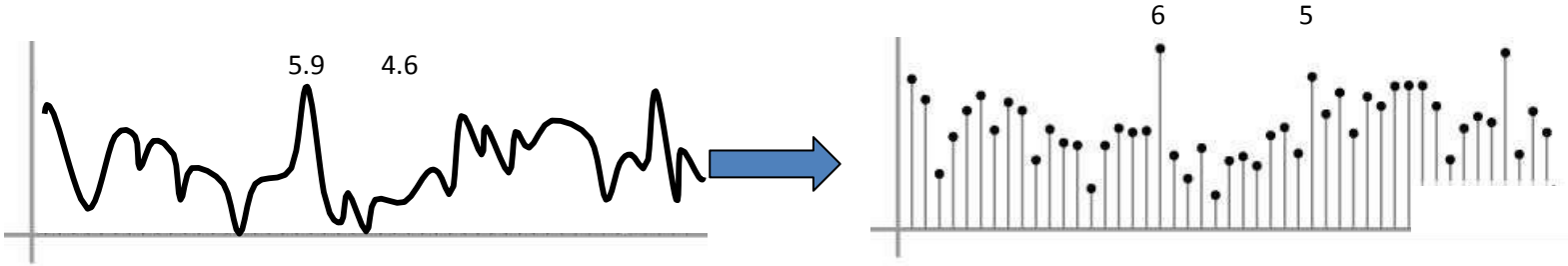
To sample this function, we take equally spaced samples along line AB, as shown in fig. 2.16 (c). The samples are shown as small white squares superimposed on the function. The set of these discrete locations gives the sampled function.

The right side of the fig. 2.16 (c) shows the gray level scale divided into eight discrete levels, ranging from black to white. The continuous gray levels are quantized simply by assigning one if the eight discrete gray levels to each sample. *The digital samples resulting from both sampling and quantization are shown in 2.16(d).*

Digital images

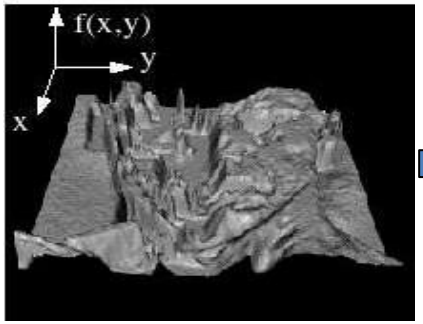
- **Quantize** each sample (round to nearest integer)
- What does quantizing signal look like?

1D



- Image thus represented as a matrix of integer values.

2D



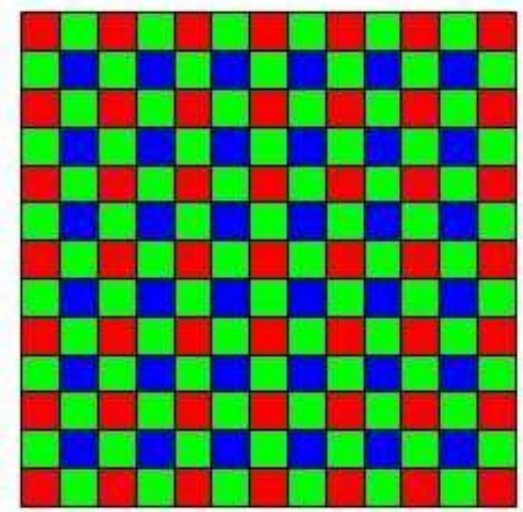
j →

i ↓

62	79	23	119	120	105	4	0
10	10	9	62	12	78	34	0
10	58	197	46	46	0	0	48
176	135	5	188	191	68	0	49
2	1	1	29	26	37	0	77
0	89	144	147	187	102	62	208
255	252	0	166	123	62	0	31
166	63	127	17	1	0	99	30

Digital color images

A (digital) **color image** is a digital **image** that includes **color** information for each pixel. A **color image** has three values (or channels) per pixel and they measure the intensity and chrominance of light



Bayer filter

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Digital color images

Color images,
RGB color space:

Split image
into three
channels

R

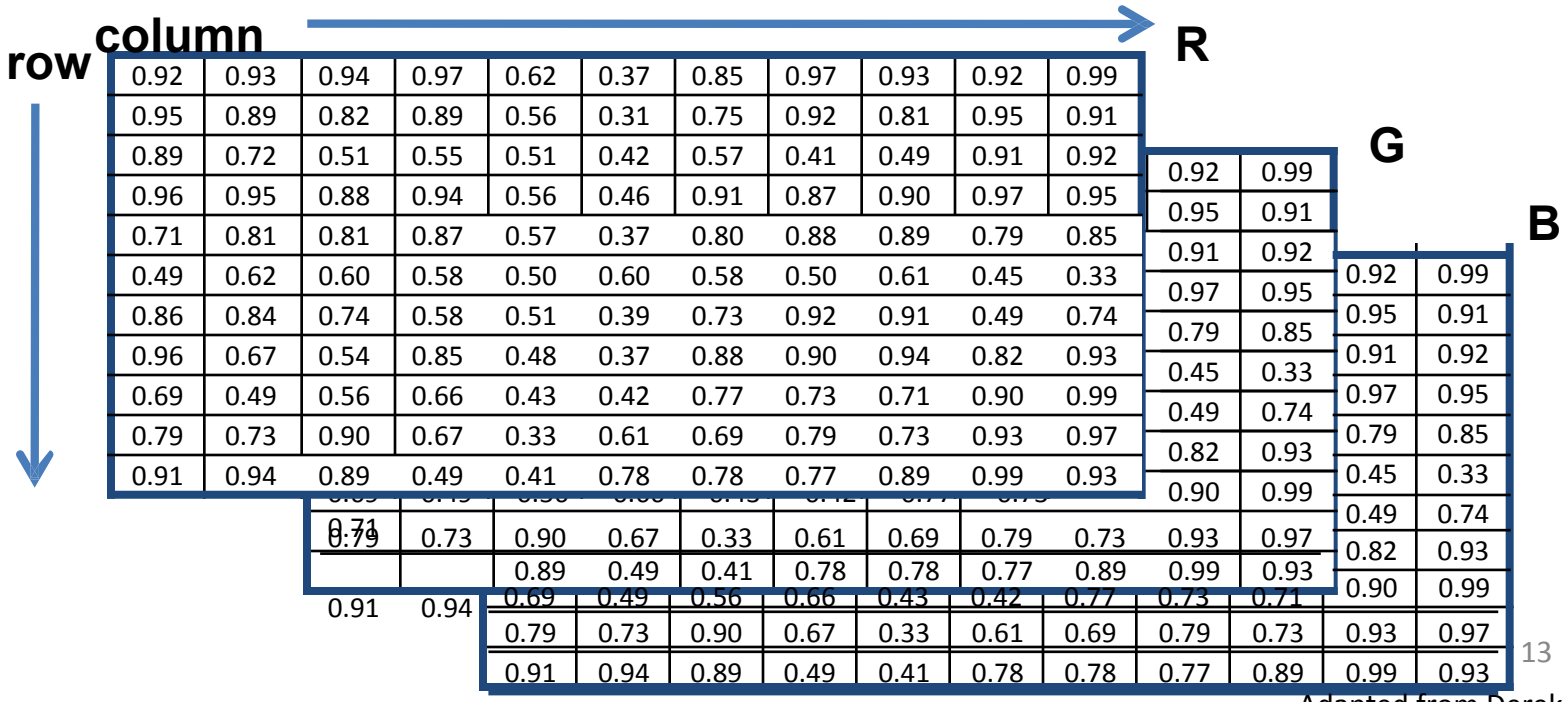
G

B

Adapted from Kristen Grauman

Images in Matlab

- Color images represented as a matrix with multiple channels (=1 if grayscale)
- Suppose we have a NxM RGB image called “im”
  - im(1,1,1) = top-left pixel value in R-channel
  - im(y, x, b) = y pixels **down**, x pixels **to right** in the bth channel
  - im(N, M, 3) = bottom-right pixel in B-channel
- imread(filename) returns a uint8 image (values 0 to 255)
  - Convert to double format with double or im2double



Adapted from Derek Hoiem